

## **SECTION IV**

### **SCOPE OF WORK**

#### **Scope of Work: Website Revamp for Kochi Metro Rail Limited (KMRL)**

##### **2.1 Project Objective**

Kochi Metro Rail Limited (KMRL) proposes to revamp its existing website into a modern, enhanced, mobile-responsive, secure, digital platform that serves as a comprehensive and citizen-centric digital hub. The objective of this initiative is to transform the current digital presence into a scalable, accessible, and future-ready ecosystem that not only showcases the Kochi Metro experience but also delivers seamless commuter services such as trip planning, fare information, and real-time alerts.

The revamped website will strengthen transparency in governance by providing structured access to

1. Provide seamless commuter services (trip planning, fare info, alerts)
2. Ensure transparency in governance (RTI, tenders, vigilance).
3. Improve user engagement and accessibility.
4. Comply with Government of India Website Guidelines (GIGW 3.0).
5. Support multilingual content (English & Malayalam).
6. Be scalable for future integrations (apps, APIs, analytics).

##### **2.2 Key Features**

###### **2.2.1 Immersive Metro Experience (Homepage)**

- Feature an engaging and immersive digital experience that reflects the essence of travel with Kochi Metro.
- A dynamic hero section will include a high-quality video capturing the journey and vibrant life around the Metro network.
- An interactive, integrated route map should display the complete network, and upon hovering over or selecting marked stations/regions, users will be able to view the next three available combined transport options, including Metro Rail, Feeder Buses, and Water Metro services.
- Incorporate nearby attractions, key landmarks, and relevant content based on the selected station.
- Seamlessly integrate live social media feeds with direct links to Kochi Metro's official social media channels.

###### **2.2.2 Ticketing and Commuter Services**

- Enable users to view fare details, check route-wise schedules, and receive service alerts.
- Prominently display monthly ridership statistics and data uploading provisions

should be provided to the concerned user departments.

- Website will provide secure and seamless redirection to official ticketing portals for online ticket purchase, recharge services and parking booking services.

### **2.2.3 Station Experience**

- Each station marker on the on the map displayed in the home page will open a detailed station profile page containing essential commuter information such as station facilities, platform details, operating hours, and connectivity options.
- Provide curated information on nearby local attractions, commercial establishments, food outlets, accessibility features (lifts, ramps, differently-abled access), parking availability, and last-mile connectivity options including feeder buses, autos, and Water Metro services.
- Localized information approach will align with the “Service Announcements” and “Metro Updates” sections outlined in the document.

### **2.2.4 Career & Recruitment Portal**

- Review and include all the functionalities in the existing portal.
- Job listings and recruitment announcements.
- Applicant login to manage profiles, upload documents, track application status, and receive updates
- Admin module with full recruitment pipeline tracking.

### **2.2.5 Informative & Feedback Services**

- Search engine related to KMRL specific data.
- Feedback and grievance modules with public participation

### **2.2.6 Sustainability & Green Metro Initiatives**

- Showcase of environmental projects and achievements
- Energy conservation efforts and carbon tracking dashboard
- Reports and certifications.

### **2.2.7 Media Library**

- Archive of event photos, campaign videos, press releases, and walkthroughs
- Search and filter by type, theme, and date
- Social sharing and download support

### **2.2.8 Additional features**

- Prominently include a dedicated link to the RTI portal to ensure transparency and ease of access to statutory disclosures, public information, and RTI-related resources.
- A structured section on Business Opportunities with Kochi Metro shall be incorporated to provide comprehensive information on commercial partnerships, advertising opportunities, leasing spaces, vendor registrations, and other collaboration prospects, thereby encouraging stakeholder participation and revenue generation.
- Clearly visible provision for Metro Parking Services shall be included, with a direct link to the official parking booking platform, enabling commuters to conveniently check availability and reserve parking spaces online.
- The vendor shall be responsible for all website copywriting in both English and Malayalam. All content must be submitted for review and will be finalized upon approval by KMRL.
- The booking engine functionality will redirect users to an external website.
- The website will be hosted on KMRL's AWS account. The vendor shall be responsible



### **For Applicants**

- Application received confirmation
- Interview invitation notifications
- Selection / rejection updates
- Email alerts
- Application Statistics and reports for HR admin

### **2.3.2 Tenders Portal**

A **Tenders Portal** is an online platform used by KMRL to publish tenders and allow vendors or contractors to engage with KMRL.

#### Key Functions

- Create and publish new tender notifications
- Upload tender documents (PDF, technical specifications, BOQ, submission guide lines)
- Edit, extend, or cancel tenders
- Manage multiple tenders from one dashboard.
- The total download count of the tender documents is to be displayed in the corresponding to the particular tender.

#### **2.3.2.1 Vendor facing Portal**

This is the **vendor/company access side** of the website.

#### Vendor Registration

- Vendors create accounts
- Submit company details and certifications
- Verify email authentication

#### Document Downloads

- View active tenders
- Download tender documents
- Check eligibility requirements
- View submission instructions

### **2.3.2.2 Post-Evaluation Publishing of Shortlisted Results**

#### Features

- Publish shortlisted bidders
- Display qualified vendors publicly
- Upload evaluation results or award notice

### **2.3.3 Vigilance Portal**

- Secure, encrypted form for public complaints or tips
- Data encrypted at rest (AES-256) and in transit (SSL)
- Access restricted to Vigilance Officers via two-factor authentication
- Backend database and export options for investigations

### **2.4. Administrative Control Panel**

for complete server maintenance, uptime, monitoring, and related support during the 365-day post go-live support period. After this period, the system will be handed over to KMRL.

- The vendor shall be responsible for conducting security audits of the website, ensuring compliance with industry standards and best practices.
- The homepage map should be implemented using static images or open-source map solutions. The use of Google Maps or any paid map services is discouraged to avoid additional costs.
- The vendor shall be responsible for the migration of data from the existing website, including but not limited to tender data and other relevant content, ensuring data integrity and completeness.

## **2.3 Internal Functional Modules**

### **2.3.1 HR/Admin Portal**

#### **Key functions:**

- Add job title, description, skills required, experience, salary, and location.
- Set application deadlines
- Publish jobs instantly on the portal
- Edit or close job postings anytime.
- Once the recruitment process completed the selected candidate is pushed to the ESS portal of KMRL through API.
- The features of the existing portal needs to be incorporated.

#### **2.3.1.1 Applicant Tracking System (ATS)**

An Applicant Tracking System manages candidates throughout the hiring process.

#### **Key functions:**

- Track applicants from application → interview → hiring
- View candidate profiles in one place
- Organize applicants by stages:
  - Applied
  - Shortlisted
  - Interview Scheduled
  - Selected / Rejected
- Recruiter notes and feedback tracking

#### **2.3.1.2 Resume Filtering**

Filters resumes based on job requirements and specify categories based on HR requirements.

#### **2.3.1.3 Status Updates and Notifications**

This feature keeps both recruiters and applicants informed throughout the hiring process.



- Role-based access for Admin, HR, Vigilance, Procurement, and Content Teams
- Drag-and-drop publishing tools (alerts, pages, FAQs, banners)
- The Admin section shall be secured through mandatory username and password authentication. Passwords must be protected using salted MD5 encryption to ensure robust security. In the event of five consecutive failed login attempts, the system shall automatically reset the administrator's password. A newly generated password will be securely transmitted to the administrator via email notification. At the database level, all passwords must be stored in encrypted format, ensuring compliance with best practices for data protection.
- Dashboards showing:
  - Commuter patterns and ticket volumes
  - Station popularity and usage
  - Feedback sentiment heat maps

## 2.5. Technical Requirements

### 2.5.1 Responsive Design

- Device compatibility: mobile, tablet, desktop, large displays
- Fully accessible interface (WCAG-compliant)

### 2.5.2 CMS and Advanced Controls

- Modular widget-based CMS for managing dynamic content
- Role-based control for secure publishing
- AI-augmented analytics (user flow, ticket trends, feedback analysis)

### 2.5.3 Security and Compliance

- End-to-end encryption (AES-256), HTTPS, OAuth2.0
- ISO 27001-aligned data protection practices
- Role-based access with MFA for sensitive modules (e.g., Vigilance)

## 2.6. Technologies Proposed

| Component | Technologies                                               |
|-----------|------------------------------------------------------------|
| Frontend  | React.js / Next.js, Bootstrap, Three.js (3D UI)            |
| Backend   | Node.js / Django / Laravel                                 |
| Database  | PostgreSQL / MongoDB                                       |
| Hosting   | AWS with CDN for global speed optimization                 |
| Security  | OAuth2.0, HTTPS, encryption at rest, ISO-aligned protocols |

## 2.7 Proposed Pages & Features

| Page Title | Key Highlights                                 |
|------------|------------------------------------------------|
| Home       | Metro simulation, alerts, banners, quick links |

|                            |                                                             |
|----------------------------|-------------------------------------------------------------|
| Ticketing & Planner        | Booking engine, fare details, AI-based journey suggestions  |
| Stations & Attractions     | Map, nearby places, real-time station feeds                 |
| Careers                    | Job listings, application portal, resume tracking           |
| Help & Feedback            | Grievance portal, user feedback interface                   |
| Sustainability Initiatives | Projects, reports, challenges, impact metrics               |
| Media Library              | Image/video archive with social features                    |
| Employee Portal            | Internal access, HR policy documents, claims, updates       |
| Tenders Portal             | Tender notices, application download/upload, result updates |
| Vigilance                  | Secure submission form, encrypted storage, access control   |

## 2.8. Implementation Timeline

| Phase                  | Duration                                                |
|------------------------|---------------------------------------------------------|
| Requirements Gathering | 10 working days                                         |
| Design & Wire framing  | 10 working days                                         |
| Development Phase      | 30 working days                                         |
| Testing & QA           | 15 working days                                         |
| Go Live & Monitoring   | 10 working days                                         |
| Total Duration:        | 75 working days (post design approval & asset delivery) |

## 2.9 Deliverables

- Fully functional, deployed website
- Admin & user documentation
- Encrypted database for sensitive modules
- Post-launch support (01 year)
- Training session (2 days included; extendable on request)

**2.10 Website Development:** Covering design, development, CMS integration, testing, deployment, and Go-Live and post implementation Support for one (1) year.